UrbanCool-AI

Chennai Urban Heat Island Mitigation Report

AI-Based Analysis of Urban Heat Islands and Optimal Tree Plantation Zones

City: Chennai, Tamil Nadu, India | Period: April–July 2025 | Resolution: 30m

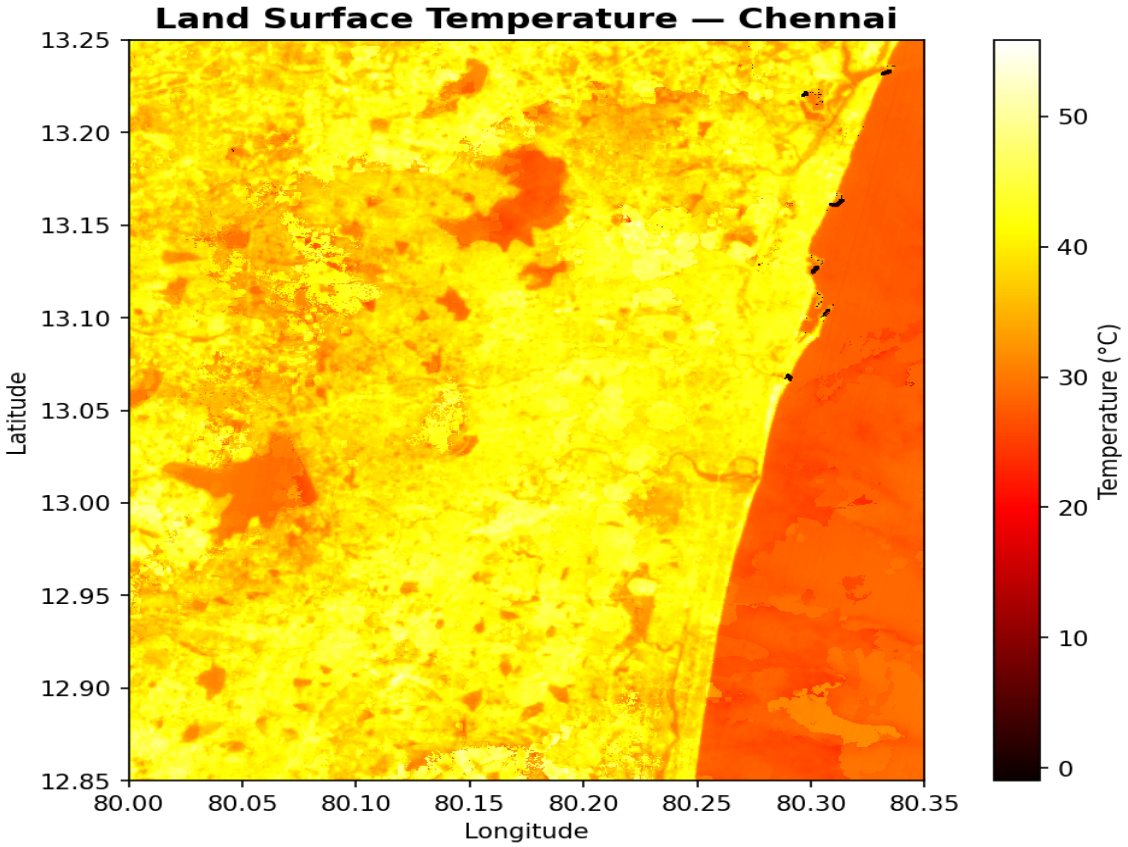
1. Executive Summary

This report presents a comprehensive analysis of Urban Heat Islands (UHI) in Chennai, India, using Landsat 8 satellite imagery from April to July 2025. The analysis identifies critical heat zones, evaluates cooling potential, and recommends optimal locations for tree plantation to mitigate urban heat effects. A machine learning model was trained to predict cooling potential, and scenario simulations demonstrate the impact of increased tree canopy coverage.

Metric	Current	After +10% Canopy	Change
Mean LST	37.52°C	36.53°C	-0.99°C
Extreme Heat Areas	27.89%	11.1%	-16.79%
Hot Areas	47.05%	32.89%	-14.16%

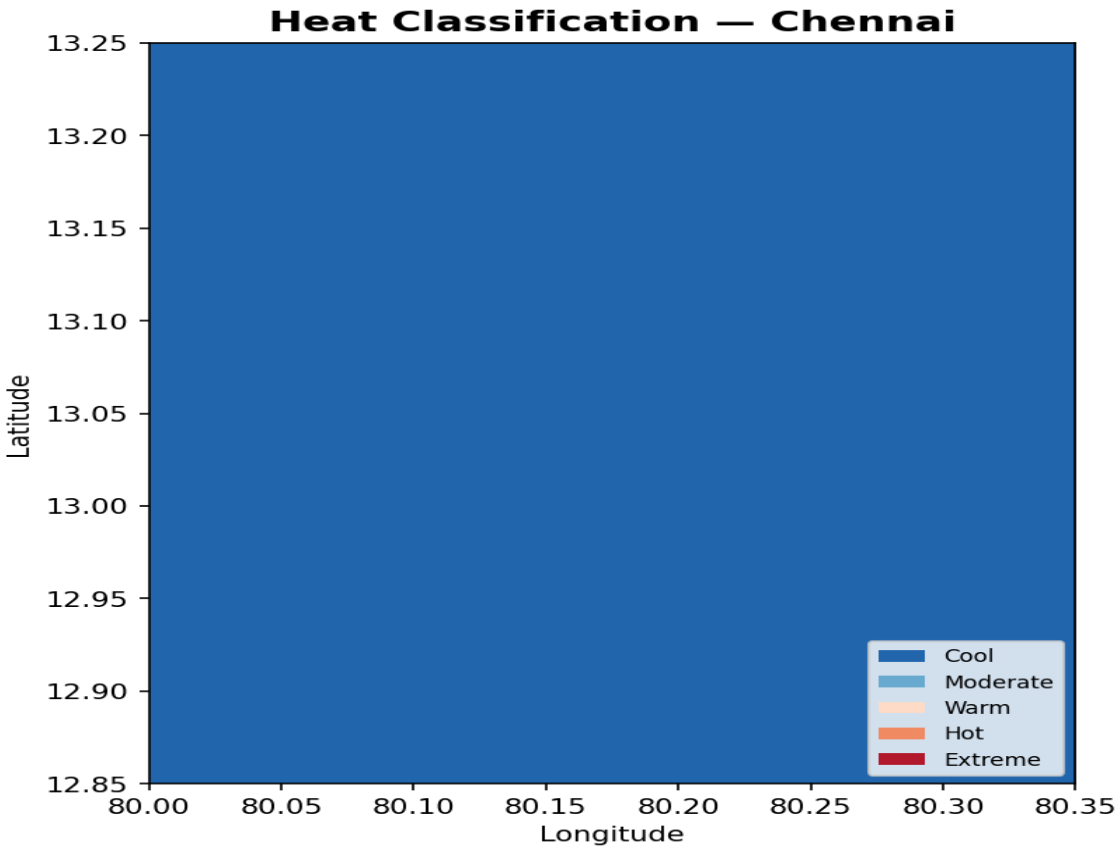
2. Land Surface Temperature

Land Surface Temperature (LST) was derived from Landsat 8 thermal band (ST_B10) and converted to Celsius. The map below shows the mean LST composite for the study period.



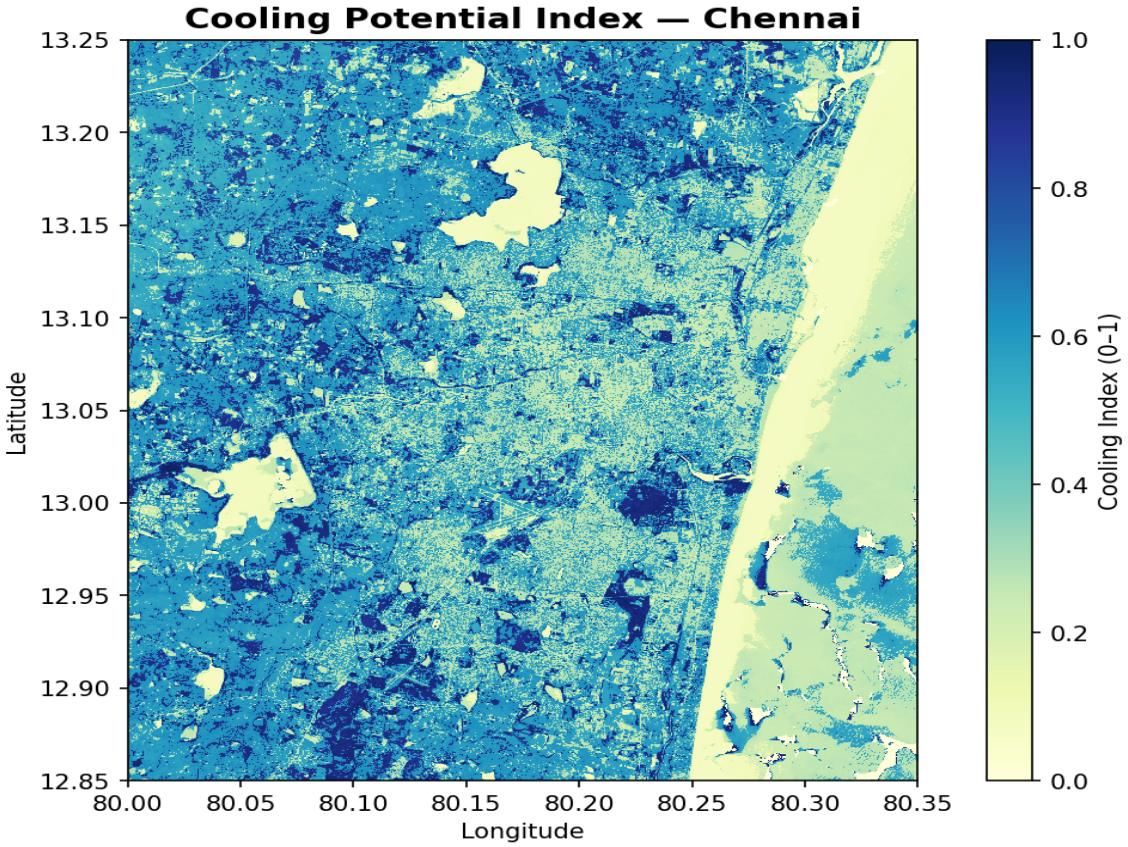
3. Heat Zone Classification

Pixels are classified relative to the mean LST: Cool (< Mean-2°C), Moderate (Mean-2°C to Mean), Warm (Mean to Mean+2°C), Hot (Mean+2°C to Mean+4°C), and Extreme (> Mean+4°C).



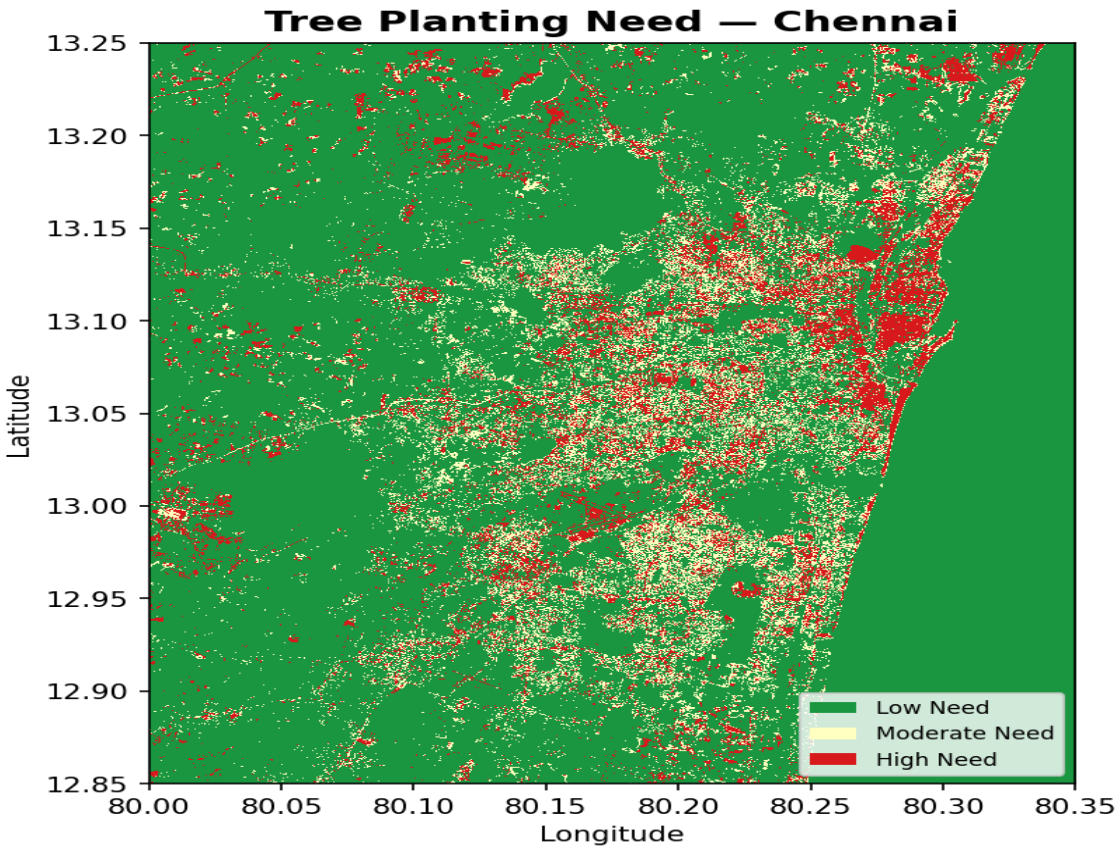
4. Cooling Potential Index

The Cooling Index combines Tree Canopy (50%), NDVI (25%), Albedo (15%), and Distance to Water (10%) into a normalized 0–1 score.



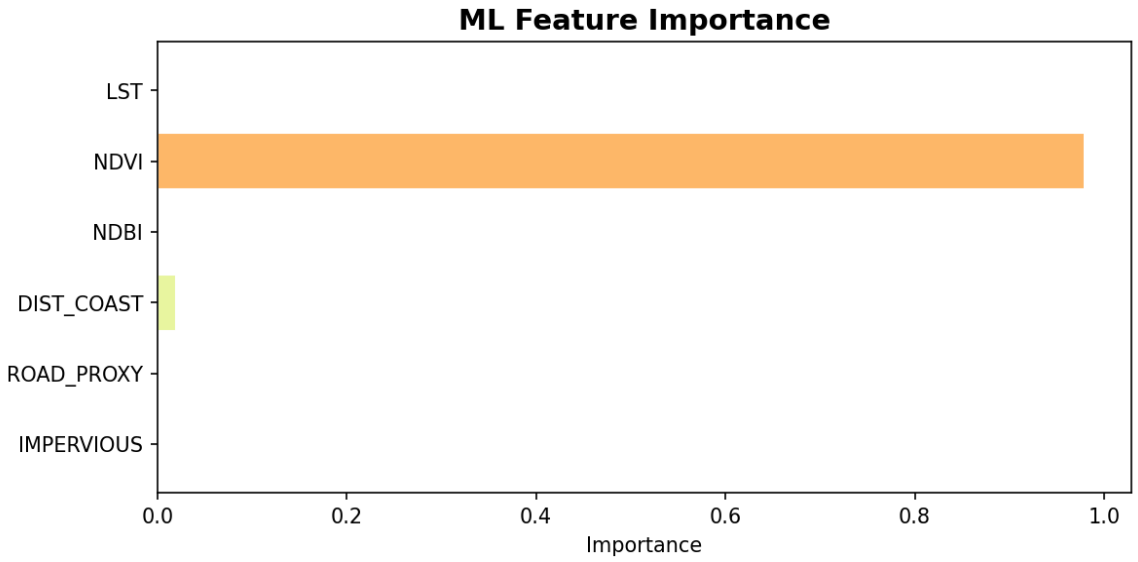
5. Tree Planting Priority Zones

Areas urgently needing trees are identified by combining hot/extreme heat classification, low NDVI (<0.3), and high impervious surface coverage.



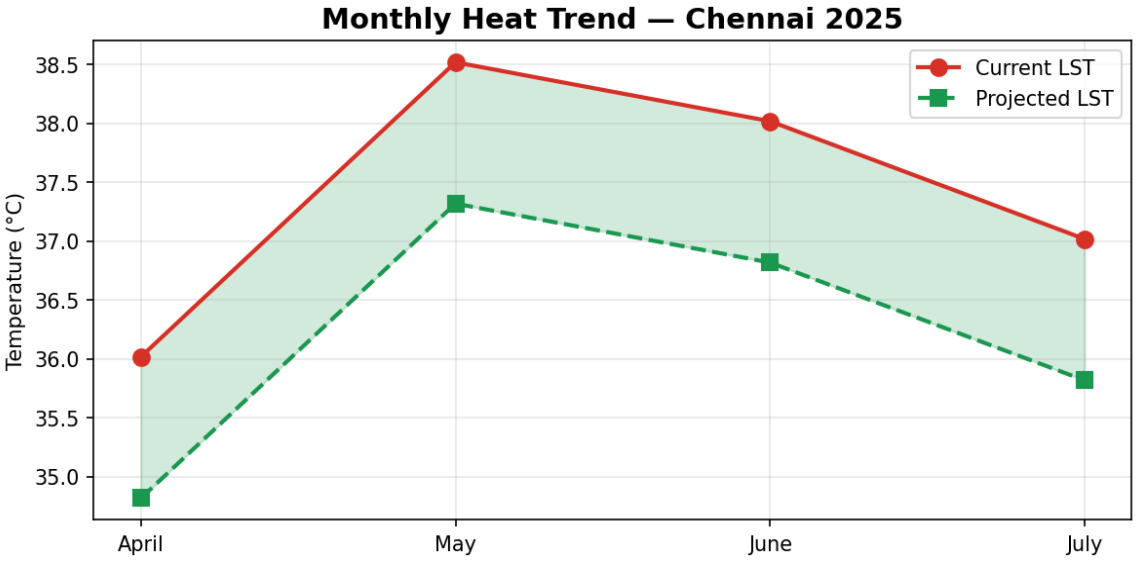
6. Machine Learning Model Performance

R² Score: 0.9996
RMSE: 0.0203
Training Samples: 3981
Test Samples: 996



7. Monthly Heat Trends & Seasonal Analysis

The monthly trend shows LST variation across the study period, with projected cooling from increased tree canopy coverage.



Historical Average: 36.0°C
Current Anomaly: 1.52°C
Interpretation: Chennai is 1.5°C above historical average. A 10% canopy increase could offset ~1.2°C of this anomaly.

8. Scenario Simulation

Scenario: +10% tree canopy increase
Average Cooling: 0.99°C
Extreme Heat Reduction: 16.79%
Hot Zone Reduction: 14.16%

— End of Report —